WEAPONS ENGINEERING TECHNICIAN EXPERIENCE QUALIFICATION 2 SUBMARINE (AKYJ)

MITE ID
123768

Jr W Eng Tech
00366-01

Curriculum Control Authority (CCA): DNP 3

MARPAC RDIMS #1426099
Approved 02 Nov 2023

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.00	Original	11 Apr 2019	D Nav P & T
2.00	See Annex I	02 Nov 2023	DNP 3

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This EQ is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Director of Naval Personnel 3 (DNP 3).

2. This EQ publication is effective on receipt and supersedes Experience Qualification 2 Version 1.0, dated 19 Apr 2011.

3. For major and minor amendments, and the approval process, contact DNP 3-2-3 or the W Eng Tech Training Manager (TM) an NTP HQ.

4. In circumstances where the EQ cannot be adhered to, submission of a Training Waiver will be required through DNP 3-2-3.

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CHAPTER 1 – GENERAL INFORMATION

AIM

1. The aim of this Experience Qualification (EQ) is to establish the requirements for attaining the AKNV EQ2 for Junior Weapons Engineering Technician 00366-01.

PREREQUISITES

2. To be selected for this course the loading authority must ensure that members meet the following prerequisites:

a. Jr W Eng Tech Applications 118937 AKNU; and

b. Jr W Eng Tech OJPR Mite Code 122837.

GRANTING OF NQUAL

3. A member will achieve the qualification W Eng Tech Maintainer OJPR AKYJ Mite Code 123768 when the standards listed in this EQ package have been met.

SUPERVISION

4. For the purpose of completing signatures within the checklist, member mentors must refer regularly to the Enabling Objectives to determine appropriate job-based activities and level of Planned Maintenance complexity. The following definitions of supervision are provided for the purposes of Junior W ENG Tech and W ENG Tech training:

5. Supervision: The Junior W ENG Tech shall be under constant supervision during their training. The degree of direct versus normal supervision shall be determined according to the level of training the Junior W ENG Tech has received in specific areas and the level of experience they may gain through the course of their EQ package.

a. It is intended that the Junior W ENG Tech shall achieve the ability to operate under normal supervision by the attainment of this EQ.

b. It is anticipated that when the Junior W ENG Tech first arrives on ship they will be under direct, continuous supervision of a qualified supervisor. An acceptable supervisor shall be deemed to be a qualified W ENG Tech as a minimum.

c. Direct Supervision shall be:

(1) Supervisor and Junior W ENG Tech shall normally be physically in the same space or under constant open communications as operational circumstances dictate. i.e. when conducting delegated tasks under one controlling authority, such as flashing up equipment; and

(2) Includes all equipment, procedural and safety related tasks.

d. Normal Supervision shall be:

(1) Physical co-location is not a requirement.

(2) Supervisor would be required to conduct spot checks to confirm performance.

(3) Supervisor will follow up to ensure the job is completed according to direction. The W ENG Tech will be under normal supervision by a W ENG Tech Maintainer. The Jr W ENG Tech is expected to accomplish all performance objectives to a proficiency level 2 and as described in the following:

Proficiency Level	Definition
1	The level of proficiency required to perform parts or elements of duties and tasks under continuous supervision.
2	The level of proficiency normally required to perform duties and tasks under normal supervision.
3	The level of proficiency required to independently and correctly perform duties and tasks.
4	The level of proficiency which usually can be acquired by considerable training and extensive practical job experience.
5	The level of proficiency indicated by a mastery of techniques and expert application procedures.
*The proficiency levels detailed above are IAW CFITES vol 3, table 9	

MAIN INSTRUCTIONS

INSTRUCTIONS FOR TRAINING PO

6. The EQ package is broken into 4 MITE codes with the NQUAL associated to the overall MITE Code.

7. The MITE Codes are listed below:

a. Overall EQ MITE Code: 123768 with NQUAL AKYJ;

b. MITE Code 123769: Annex A - W Eng Tech Communications;

c. MITE Code 123770: Annex B – W Eng Tech Fire Control; and

d. MITE Code 123771Annex C - W Eng Tech SONAR.

8. The member will be assigned to a W Eng Tech Sub occ by the Career Manager on completion of Experience Qualification 1 (EQ1) when granted the Mite Code 122837. In the EQ2 package, the member will be required to complete the Annex associated with their new assigned W Eng Tech Sub Occupation and the overall EQ2 package in MMTT.

9. In MMTT there is the overall EQ2 package and a package for each Annex. Once the member has completed the assigned Annex and the training PO has completed the annex in MMTT, the overall package must be completed in according with instructions in the package.

10. Contact the W Eng Tech Training Manager if unclear on instructions.

COMPLETION TIME AND EXTENSIONS

11. The time to complete this EQ is no less than 6 months and a maximum of 12 months. If necessary, the Member's Commanding Officer (CO) may grant an initial extension of 3 months. If a second extension is required, DNP shall be the approving authority. On the commencement of the initial extension:

a. The member's unit to send Initial Extension message (template below);
b. The member's unit to commence RM's IAW DAOD 5019; and
c. The member's unit to ensure feedback notes are generated.

Note: Unit should consider initiating RM when it is identified that, following the current progression rate or for other reasons, the member would unlikely be successful within the allotted time.

12. If the member has not completed the EQ package on completion of the initial extension, the member's division CPO is to schedule a meeting to make a recommendation to the Occupation Manager (OM) DNP 3-2-3 on how to proceed with the member.

13. The meeting members is to include: (from associated coast)

a. NFS CPO2;
b. Fleet CSEC;
c. Coastal Advisor; and
d. Members Div CPO2.

14. Recommendations from the meeting to the OM could be:

a. To have an Administrative Review (AR) conducted on the member;
b. To have a second extension; or
c. To have a suspension of training (include reason).

15. If the decision is to conduct an AR, the OM will inform the CM who will initiate the AR. The possible outcome of the AR could be:

a. Recommend extension and continue with training;
b. Occupational transfer;
c. Transfer between sub-components;
d. An offer of terms of service in any case in which an offer has not been made by CAF authorities; or
e. Release or recommendation for release, as applicable.

16. If the recommendation is to request a second extension:

a. The member's unit is to send the second extension request message (template below);
b. The member's unit to continue with RM's IAW DAOD 5019; and
a. The member's unit to ensure feedback notes are generated.

17. If the recommendation is a suspension of training:

a. The member's unit is to send an encrypted email to the OM explaining the situation;
b. If applicable, member's unit to continue with RM's IAW DAOD 5019; and
c. The member's unit to ensure feedback notes are generated.

18. At all times the member is to continue to complete the EQ package.

Initial Extension Template

TO CANFLTPAC HQ ESQUIMALT//F4 ENG//
CANFLTLANT HQ HALIFAX//F4 ENG//
INFO NPTG HQ ESQUIMALT// W ENG TECH TM//
NDHQ DGMC OTTAWA// D MIL C 2-4-3//
NDHQ C NAVY DNP OTTAWA//DNP 3-2-3//
PCC ESQUIMALT// W ENG TECH OA //
PCC HALIFAX// W ENG TECH OA //
SIC WAF
SUBJ: EQ INITIAL EXTENSION (EQ NAME, MITE CODE)
1. (SN, RANK, NAME, INIT, MOSID NUMBER, MOS TITLE) HAS BEEN GRANTED AN INITIAL EXTENSION OF (3 MONTHS MAX) FOR EQ (NAME) COMPLETION
2. (REASON FOR EXTENSION)
3. UNIT TRAINING COORDINATOR IS TO UPDATE MBRS MMTT
4. POC: (UNIT TRAINING COORDINATOR)

Second Extension Request Template

TO NDHQ C NAVY DNP OTTAWA//DNP 3-2-3//
INFO CANFLTPAC HQ ESQUIMALT//F4 ENG//
CANFLTLANT HQ HALIFAX//F4 ENG//
NPTG HQ ESQUIMALT// W ENG TECH TM//
NDHQ DGMC OTTAWA// D MIL C 2-4-3//
PCC ESQUIMALT// W ENG TECH OA //
PCC HALIFAX// W ENG TECH OA //
SIC WAF
SUBJ: EQ SECOND EXTENSION REQUEST (EQ NAME, MITE CODE)
1. (SN, RANK, NAME, INIT, MOSID NUMBER, MOS TITLE) IS REQUESTING A SECOND EXTENSION FOR EQ (NAME) COMPLETION
2. (RECOMMENDATION FOR EXTENSION)
3. UNIT TRAINING COORDINATOR IS TO UPDATE MBRS MMTT
4. POC: (UNIT TRAINING COORDINATOR)

Annex A - Weapons Engineering Technician Submarine Communications

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Annex A - EQ2 W Eng Tech Submarine Communications

General Instructions

1. One copy of this EQ Annex will be provided to the member and one copy of Annex hereinafter referred to as the master copy, shall be made up for each individual beginning OJT and be retained by the Departmental Training Petty Officer.

2. A divisional interview shall be conducted to officially start the timeline and outline the process to the member. This is also an opportunity for the supervisor to agree to provide the member time to progress and complete training. The Unit / Member Agreement Form, at the end of these instructions, is to be signed during the divisional interview and kept with the Division Training PO. A copy is to be kept in the Annex as a reference to the member.

3. The tables of contents below include a list of equipment for you to gain experience on to make to prepared to become a Trade Journeyman. You will be required to identify the major components of the systems, how to operate the systems, perform Maintenance, and Perform a Mini Board on the system.

4. Each system below will have associated tables. On the right side of the tables are the date completed and Name and Signature blocks. The Name and Signature will be from the senior section technician or a W Eng Tech PO1. The PO's do not have to be completed in order and Name and Signatures can be granted through out the period. After the Name and Signatures are completed for each PO, a mini board is to be conducted and chaired by the CHOD or their designated PO1.

5. The development of the Member through the EQ process is critical to the creation of skilled and experienced technicians. To this end, the Name and Signature of the supervisors indicating completion of EOs and POs must reflect a high standard of accomplishment by the Member;

6. If deemed essential by the Sr Technician or Manager, the Member may have to perform more than the required number of times specified IOT satisfactorily complete the EO/PO at the high level of proficiency expected and get the Name and Signature.

Mini-Board Instructions

1. The board is made up of:

   a. CSED CHOD (scoring);

   b. SECTION HEAD (scoring);

   c. Subject Matter Expert (non-scoring member); and

2. Material and facilities requirement:

   a. Pens, pencils and stationary;

b. Specified References;

c. Print out of marking guide (Annex G) for each member of the scoring board members; and

d. Suitable location.

3. Board Procedure:

a. Board members will meet with the member to ensure all supplies, references and equipment is available. They will explain the board procedures.

b. The member is then given 45 minutes before the Oral Board commencement to prepare the drawing as fitted in the member's ship.

c. The drawing must include:

i. Major Components;

ii. Major interconnections;

iii. Power Requirements; and

iv. Environmental requirements.

d. The member is complete the Oral Presentation in 25 Minutes by:

i. Describing the Purpose of system;

ii. Identify and explain major components;

iii. Identify and explain major signals flow; Identify and explain power requirements;

iv. Identify and explain environmental requirements;

v. Identify and talk about safety requirements; and

vi. Answer board questions.

4. Following the board, the member will be debriefed on their score, boardman-ship, strength, and areas for development.

<u>Final Board Instructions</u>

1. When the member has completed the associated Annex, the member can prepare for and then complete the Final Board. Details and instructions for the final board can be found at Annex F Final Board.

Annex A - EQ2 W Eng Tech Submarine Communications

Administration Instructions

1. The time to complete and instruction for extensions are in the main instructions of the EQ.

2. Members are to be cautioned that failure to complete the EQ, including requests for Cease Training, may result in an Administrative Review IAW NPTORD 4500-29 Course Non-Concurrence, Deferrals and Removals.

3. As the member completes each EO and PO to the required standard, the Training PO shall record the progress of the member in MMTT, provide the Member with a copy and report the progress to the CHOD on a Monthly basis. The Progression Review Table is to be signed monthly by the Division Training PO and kept in their file with the Member Agreement Form.

4. It may not be possible to conduct some EOs/POs. If, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable, therefore:

   a. The Departmental Training PO shall arrange an alternate unit/training establishment to conduct the EO/PO. Maintenance Procedural Trainer (MPT) may be utilized as a last resort if no miscellaneous option is available within the timeframe of the EQ.

5. Name and Signs may not be required under some PO/EO's. Some EO's will not apply, and N/A can be written into the Name and Signature block. For example, N/A can be written into the Confined Space Name and Signature block under Cranes.

Annex A - EQ2 W Eng Tech Submarine Communications

Unit / Member Agreement Form

Name and Initial: _____

Rank: _____

SN: _____

Date Training Started: _____

Expected Date of Completion: _____

Divisional Interview Conducted by:

DivO: _____

Supervisor: _____

Note: THE EQ MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE EQ WILL BE RETURNED FOR RE-SUBMISSION.

Annex A - EQ2 W Eng Tech Submarine Communications

PROGRESSION REVIEW

Month	EQ REVIEWED BY (RANK, POSITION, NAME)	SIGNATURE	DATE
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
First Extension (Unit CO)			
Second Extension (DNP 3)			

Annex A - EQ2 W Eng Tech Submarine Communications

Table of contents

PO 001 Internal and Other Communications
EO 001.01 VCS 446, Maintenance, and Command lines
EO 001.01.01 Identify the VCS 446, Maintenance, and Command lines Major Components
EO 001.01.02 Operate the VCS 446, Maintenance, and Command lines
EO 001.01.03 Perform Maintenance on VCS 446, Maintenance, and Command lines
EO 001.02 Shore lines
EO 001.02.01 Identify the Shore lines Major Components
EO 001.02.02 Operate the Shore lines
EO 001.02.03 Perform Maintenance on Shore lines
EO 001.03 Victoria internal communications system (VICES)
EO 001.03.01 Identify the VICES Major Components
EO 001.03.02 Operate the VICES
EO 001.03.03 Perform Maintenance on VICES
EO 001.04 ER/MR Wireless System
EO 001.04.01 Identify the ER/MR Wireless System Major Components
EO 001.04.02 Operate the ER/MR Wireless System
EO 001.04.03 Perform Maintenance on ER/MR Wireless System
EO 001.05 Escape Communications
EO 001.05.01 Identify the Escape Communications Major Components
EO 001.05.02 Operate the Escape Communications
EO 001.05.03 Perform Maintenance on Escape Communications
EO 001.06 Perform Mini Board on Internal and Other Communications

PO 002 VLF Systems
EO 002.01 Buoyant Wire Ariel (BWA)
EO 002.01.01 Identify the BWA Major Components
EO 002.01.02 Operate the BWA
EO 002.01.03 Perform Maintenance on BWA
EO 002.02 VHF / AHG
EO 002.02.01 Identify the VHF / AHG Major Components
EO 002.02.02 Operate the VHF / AHG
EO 002.02.03 Perform Maintenance on VHF / AHG
EO 003.03 ASU, TSU, & Patch Panels
EO 002.03.01 Identify the ASU, TSU, & Patch Panels Major Components
EO 002.03.02 Operate the ASU, TSU, & Patch Panels
EO 002.03.02 Perform Maintenance on ASU, TSU, & Patch Panels
EO 002.04 Perform Mini Board on VLF Systems

PO 003 HF, VHF, and UHF Communications
EO 003.01 HF
EO 003.01.01 Identify the HF Major Components
EO 003.01.02 Operate the HF
EO 003.01.03 Perform Maintenance on HF

Annex A - EQ2 W Eng Tech Submarine Communications

EO 003.02 VHF
EO 003.02.01 Identify the VHF Major Components
EO 003.02.02 Operate the VHF
EO 003.02.03 Perform Maintenance on VHF
EO 003.03 UHF
EO 003.03.01 Identify the UHF Major Components
EO 003.03.02 Operate the UHF
EO 003.03.03 Perform Maintenance on UHF
EO 003.04 Communications Mast
EO 003.04.01 Identify the Communications Mast Major Components
EO 003.04.02 Operate the Communications Mast
EO 003.04.03 Perform Maintenance on Communications Mast
EO 003.05 Perform Mini Board on HF, VHF, and UHF Communications

PO 004 Protected MILSATCOM Systems
EO 004.01 Universal Modular Mast
EO 004.01.01 Identify the Universal Modular Mast Major Components
EO 004.01.02 Operate the Universal Modular Mast
EO 004.01.03 Perform Maintenance on Universal Modular Mast
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EO 004.02.01 Identify the NMT Major Components
EO 004.02.02 Operate the NMT
EO 004.02.03 Perform Maintenance on NMT
EO 004.03 Perform Mini Board on Protected MILSATCOM Systems

PO 005 RADAR Suite
EO 005.01 Identification Friend or Foe (IFF)
EO 005.01.01 Identify the IFF Major Components
EO 005.01.02 Operate the IFF
EO 005.01.03 Perform Maintenance on IFF
EO 005.02 Electronic Warfare (EW)
EO 005.02.01 Identify the EW System Major Components
EO 005.02.02 Operate the EW System
EO 005.02.03 Perform Maintenance on EW System
EO 005.03 RADAR
EO 005.03.01 Identify the RADAR Major Components
EO 005.03.02 Operate the RADAR
EO 005.03.03 Perform Maintenance on RADAR
EO 005.04 Perform Mini Board on RADAR Suite

PO 006 ERT and Bulkhead 35 Duties
EO 006.01 Perform as a member of the ERT
EO 006.02 Perform as the 35 Bulkhead Watch Keeper

PO 007 Final Board

Annex A - EQ2 W Eng Tech Submarine Communications

PO 001 Internal and Other Communications			
EO 001.01.01 Identify the VCS 446, Maintenance, and Command lines Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Garde 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.01.02 Operate the VCS 446, Maintenance, and Command lines		Date	Signature
	Perform 48 hour pre-sails		

Annex A - EQ2 W Eng Tech Submarine Communications

Operate the VCS 446	Perform Flash up		
	Verify operational		
	Brief Senior Hand		
EO 001.01.03 Perform Maintenance on VCS 446, Maintenance, and Command lines		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

PO 001 Internal and Other Communications			
EO 001.02.01 Identify the Shore lines Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.02.02 Operate the Shore lines		Date	Signature

Annex A - EQ2 W Eng Tech Submarine Communications

Operate the Shore lines	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 001.02.03 Perform Maintenance on Shore lines		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

PO 001 Internal and Other Communications			
EO 001.03.01 Identify the Victoria internal communications system (VICES) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.03.02 Operate the VICES		Date	Signature
	Perform 48 hour pre-sails		

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Operate the VICES	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 001.03.03 Perform Maintenance on VICES		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM Routine		
	Assist with 3 Monthly PM Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

PO 001 Internal and Other Communications			
EO 001.04.01 Identify the ER/MR Wireless System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

Annex A - EQ2 W Eng Tech Submarine Communications

EO 001.04.02 Operate the ER/MR Wireless System		Date	Signature
Operate the ER/MR Wireless System	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 001.04.03 Perform Maintenance on ER/MR Wireless System		**Date**	**Signature**
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

PO 001 Internal and Other Communications			
EO 001.05.01 Identify the Escape Communications Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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Annex A - EQ2 W Eng Tech Submarine Communications

EO 001.05.02 Operate the Escape Communications		Date	Signature
Operate the Escape comms	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 001.05.03 Perform Maintenance on Escape Communications		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

EO 001.06 Perform Mini Board on Internal and Other Communications			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

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PO 002 VLF Systems			
EO 002.01.01 Identify the Buoyant Wire Ariel (BWA) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

Annex A - EQ2 W Eng Tech Submarine Communications

EO 002.01.02 Operate the BWA		Date	Signature
Operate the BWA	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 002.01.03 Perform Maintenance on BWA		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

PO 002 VLF Systems			
EO 002.02.01 Identify the VHF / AHG Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.02.02 Operate the VHF / AHG		Date	Signature

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Operate the VHF / AHG	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 002.02.03 Perform Maintenance on VHF / AHG		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 002 VLF Systems			
EO 002.03.01 Identify the ASU, TSU, & Patch Panels Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 002.03.02 Operate the ASU, TSU, & Patch Panels		Date	Signature
Operate the ASU, TSU & Patch Panels	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 002.03.03 Perform on ASU, TSU, & Patch Panels		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

EO 002.04 Perform Mini Board on VLF Systems			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

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PO 003 HF, VHF, and UHF Communications			
EO 003.01.01 Identify the HF Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 003.01.02 Operate the HF		Date	Signature
Operate the HF	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.01.03 Perform Maintenance on HF		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

PO 003 HF, VHF, and UHF Communications			
EO 003.02.01 Identify the VHF Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 003.02.02 Operate the VHF		Date	Signature
Operate the VHF	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.02.03 Perform Maintenance on VHF		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 003 HF, VHF, and UHF Communications			
EO 003.03.01 Identify the UHF Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 003.03.02 Operate the UHF		Date	Signature
Operate the UHF	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.03.03 Perform Maintenance on UHF		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 003 HF, VHF, and UHF Communications			
EO 003.04.01 Identify the Communications Mast Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 003.04.02 Operate the Communications Mast		Date	Signature
Operate the Comms Mast	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.04.03 Perform Maintenance on Communications Mast		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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EO 003.05 Perform Mini Board on HF, VHF, and UHF Communications			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

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PO 004 Protected MILSATCOM Systems			
EO 004.01.01 Identify the Universal Modular Mast Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 004.01.02 Operate the Universal Modular Mast		Date	Signature
Operate the Universal Modular Mast	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 004.01.03 Perform Maintenance on Universal Modular Mast		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 004 Protected MILSATCOM Systems			
EO 004.02.01 Identify the Naval Maintenance Terminal (NMT) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 004.02.02 Operate the NMT		Date	Signature
Operate the NMT	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 004.02.03 Perform Maintenance NMT		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

EO 004.04 Perform Mini Board on Protected MILSATCOM Systems			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

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PO 005 RADAR Suite			
EO 005.01.01 Identify the IFF Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 005.01.02 Operate the IFF		Date	Signature
Operate the IFF	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 005.01.03 Perform Maintenance on IFF		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 005 RADAR Suite			
EO 005.02.01 Identify the EW System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 005.02.02 Operate the EW System		Date	Signature

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Operate the EW System	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 005.02.03 Perform Maintenance on EW System		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 005 RADAR Suite			
EO 005.03.01 Identify the RADAR Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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EO 005.03.02 Operate the RADAR		Date	Signature
Operate the RADAR	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 005.03.03 Perform Maintenance on RADAR		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Assist with Daily PM Routine		
	Assist with Weekly PM Routine		
	Assist with Monthly PM's Routine		
Assist with Major PM	Assist with 6 Monthly PM Routine		
	Assist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex A - EQ2 W Eng Tech Submarine Communications

<table>
<tr><th colspan="4">EO 005.04 mini board on RADAR Suite</th></tr>
<tr><td rowspan="5">Draw system block diagram to include</td><td>Major Components</td><td>Date Completed</td><td>Signature</td></tr>
<tr><td>Major Signals Flow</td><td></td><td></td></tr>
<tr><td>Power requirements</td><td colspan="2">Board Member Chair name</td></tr>
<tr><td>Environmental requirements</td><td colspan="2" rowspan="2"></td></tr>
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<tr><td rowspan="7">Explain / talk about</td><td>Purpose of system</td><td colspan="2">Overall Board Comments</td></tr>
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<tr><td>Identify and explain environmental requirements</td></tr>
<tr><td>Identify and talk about safety requirements</td></tr>
</table>

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Annex A - EQ2 W Eng Tech Submarine Communications

PO 006 ERT and Bulkhead 35 Duties			
EO 006.01 Perform as a member of the ERT	Interpreting technical references	Date	Name and Sign
	Perform IAW HQ1 direction	Date	Name and Sign
	Clarify reports to HQ1	Date	Name and Sign
EO 006.02 Perform as the 35 Bulkhead Watch Keeper	Operating systems forward of 35 bulkhead	Date	Name and Sign
	Conducting rounds forward of 35 bulkhead	Date	Name and Sign
	Performing Emergency Operating Procedures (EOPs) as required	Date	Name and Sign

Annex A - EQ2 W Eng Tech Submarine Communications

PO 007 Complete Final Board and Completion of Admin			
EO 006.01 Final Board	Section 1 Part A	Date	Name and Sign
	Section 1 Part B	Date	Name and Sign
	Section 2 Part A	Date	Name and Sign
	Section 2 Part B	Date	Name and Sign
	MMTT Completed	Date	Name and Sign
EO 006.02 Completion Cert	Completion Certificate Annex H Completed	Date	Name and Sign
EO 006.03	Complete Overall EQ Package in MMTT	Date	Name and Sign
EO 006.04 Message Released	Message Released Annex J Completed	Date	Name And Sign

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Annex B – Weapons Engineering Technician Submarine Fire Control

MARPAC RDIMS #1426099
Approved 02 Nov 2023

Annex B - EQ2 W Eng Tech Submarine Fire Control

Introduction / Instructions

General Instructions

7. One copy of this EQ Annex will be provided to the member and one copy of Annex hereinafter referred to as the master copy, shall be made up for each individual beginning OJT and be retained by the Departmental Training Petty Officer.

8. A divisional interview shall be conducted to officially start the timeline and outline the process to the member. This is also an opportunity for the supervisor to agree to provide the member time to progress and complete training. The Unit / Member Agreement Form, at the end of these instructions, is to be signed during the divisional interview and kept with the Division Training PO. A copy is to be kept in the Annex as a reference to the member.

9. The tables of contents below include a list of equipment for you to gain experience on to make to prepared to become a Trade Journeyman. You will be required to identify the major components of the systems, how to operate the systems, perform Maintenance, and Perform a Mini Board on the system.

10. Each system below will have associated tables. On the right side of the tables are the date completed and Name and Signature blocks. The Name and Signature will be from the senior section technician or a W Eng Tech PO1. The PO's do not have to be completed in order and Name and Signatures can be granted through out the period. After the Name and Signature are completed for each PO, a mini board is to be conducted and chaired by the CHOD or their designated PO1.

11. The development of the Member through the EQ process is critical to the creation of skilled and experienced technicians. To this end, the Name and Signature of the supervisors indicating completion of EOs and POs must reflect a high standard of accomplishment by the Member;

12. If deemed essential by the Sr Technician or Manager, the Member may have to perform more than the required number of times specified IOT satisfactorily complete the EO/PO at the high level of proficiency expected and get the Name and Signature.

Mini-Board Instructions

5. The board is made up of:

   a. CSED CHOD (scoring);

   b. SECTION HEAD (scoring);

   c. Subject Matter Expert (non-scoring member); and

6. Material and facilities requirement:

a. Pens, pencils and stationary;

b. Specified References;

c. Print out of marking guide (Annex G) for each member of the scoring board members; and

d. Suitable location.

7. Board Procedure:

a. Board members will meet with the member to ensure all supplies, references and equipment is available. They will explain the board procedures.

b. The member is then given 45 minutes before the Oral Board commencement to prepare the drawing as fitted in the member's ship.

c. The drawing must include:

i. Major Components;

ii. Major interconnections;

iii. Power Requirements; and

iv. Environmental requirements.

d. The member is complete the Oral Presentation in 25 Minutes by:

i. Describing the Purpose of system;

ii. Identify and explain major components;

iii. Identify and explain major signals flow; Identify and explain power requirements;

iv. Identify and explain environmental requirements;

v. Identify and talk about safety requirements; and

vi. Answer board questions.

8. Following the board, the member will be debriefed on their score, boardman-ship, strength, and areas for development.

<u>Final Board Instructions</u>

Annex B - EQ2 W Eng Tech Submarine Fire Control

2. When the member has completed the associated Annex, the member can prepare for and then complete the Final Board. Details and instructions for the final board can be found at Annex F Final Board.

Administration Instructions

6. The time to complete and instruction for extensions are in the main instructions of the EQ.

7. Members are to be cautioned that failure to complete the EQ, including requests for Cease Training, may result in an Administrative Review IAW NPTORD 4500-29 Course Non-Concurrence, Deferrals and Removals.

8. As the member completes each EO and PO to the required standard, the Training PO shall record the progress of the member in MM, provide the Member with a copy and report the progress to the CHOD on a Monthly basis. The Progression Review Table is to be signed monthly by the Division Training PO and kept in their file with the Member Agreement Form.

9. It may not be possible to conduct some EOs/POs. If, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable, therefore:

b. The Departmental Training PO shall arrange an alternate unit/training establishment to conduct the EO/PO. Maintenance Procedural Trainer (MPT) may be utilized as a last resort if no miscellaneous option is available within the timeframe of the EQ.

10. Name and Signs may not be required under some PO/EO's. Some EO's will not apply, and N/A can be written into the Name and Sign block. For example, N/A can be written into the Confined Space Name and Signature block under Cranes.

Annex B - EQ2 W Eng Tech Submarine Fire Control

Unit / Member Agreement Form

Name and Initial: _____

Rank: _____

SN: _____

Date Training Started: _____

Expected Date of Completion: _____

Divisional Interview Conducted by:

DivO: _____

Supervisor: _____

Note: THE EQ MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE EQ WILL BE RETURNED FOR RE-SUBMISSION.

Annex B - EQ2 W Eng Tech Submarine Fire Control

PROGRESSION REVIEW

Month	EQ REVIEWED BY (RANK, POSITION, NAME)	SIGNATURE	DATE
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
First Extension (Unit CO)			
Second Extension (DNP 3)			

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Annex B - EQ2 W Eng Tech Submarine Fire Control

Table of Contents

PO 001 Weapons Handling System
EO 001.01 Identify the Weapons Handling System Major Components
EO 001.02 Operate the Weapons Handling System
EO 001.03 Perform maintenance on Weapons Handling System
EO 001.04 Perform Mini Board on Weapons Handling System

PO 002 Weapons Discharge System
EO 002.01.01 Identify the Weapons Discharge System Major Components
EO 002.01.02 Operate the Weapons Discharge System
EO 002.01.03 Perform maintenance on Weapons Discharge System
EO 002.03 Perform Mini Board on Weapons Discharge System

PO 003 Submerged Signal Ejector
EO 003.01.01 Identify the Submerged Signal Ejector Major Components
EO 003.01.02 Operate the Submerged Signal Ejector
EO 003.01.03 Perform maintenance on Submerged Signal Ejector
EO 003.03 Perform Mini Board on Submerged Signal Ejector

PO 004 ERT and Bulkhead 35 Duties
EO 004.01 Perform as a member of the ERT
EO 004.02 Perform as the 35 Bulkhead Watch Keeper

EO 005 Final Board

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Annex B - EQ2 W Eng Tech Submarine Fire Control

PO 001 Weapons Handling System			
EO 001.01 Identify the Weapons Handling System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		

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Annex B - EQ2 W Eng Tech Submarine Fire Control

EO 001.02 Operate the Weapons Handling System		Date	Signature
Operate the Weapons Handling System	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief TI		
EO 001.03 Perform maintenance on Weapons Handling System		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM's Routine		
	Asist with Monthly PM's Routine		
Perform Visual Inspection	Embarkation & hoist Structure		
	Traversing & Section Headage		
	Loading Structure		
	Hydraulic System		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex B - EQ2 W Eng Tech Submarine Fire Control

<table>
<tr><th colspan="4">EO 001.04 mini board on Weapons Handling System</th></tr>
<tr><td rowspan="5">Draw system block diagram to include</td><td>Major Components</td><td>Date Completed</td><td>Signature</td></tr>
<tr><td>Major Signals Flow</td><td></td><td></td></tr>
<tr><td>Power requirements</td><td colspan="2">Board Member Chair name</td></tr>
<tr><td>Environmental requirements</td><td colspan="2" rowspan="2"></td></tr>
<tr><td>Environmental requirements</td></tr>
<tr><td rowspan="7">Explain / talk about</td><td>Purpose of system</td><td colspan="2">Overall Board Comments</td></tr>
<tr><td>Identify and explain major components</td><td colspan="2" rowspan="6"></td></tr>
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<tr><td>Identify and explain environmental requirements</td></tr>
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Annex B - EQ2 W Eng Tech Submarine Fire Control

PO 002 Weapons Discharge System			
EO 002.01.01 Identify the Weapons Discharge System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.01.02 Operate the Weapons Discharge System		Date	Signature
Operate the WD	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief TI		

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Annex B - EQ2 W Eng Tech Submarine Fire Control

EO 002.01.03 Perform maintenance on Weapons Discharge System		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

N Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex B - EQ2 W Eng Tech Submarine Fire Control

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Annex B - EQ2 W Eng Tech Submarine Fire Control

PO 003 Submerged Signal Ejector			
EO 003.01.01 Identify the Submerged Signal Ejector Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.01.02 Operate the Submerged Signal Ejector		Date	Signature
Operate the SSE	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief TI		

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Annex B - EQ2 W Eng Tech Submarine Fire Control

EO 003.01.03 Perform maintenance on Submerged Signal Ejector		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex B - EQ2 W Eng Tech Submarine Fire Control

EO 003.04 mini board on Submerged Signal Ejector			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

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Annex B - EQ2 W Eng Tech Submarine Fire Control

PO 004 ERT and Bulkhead 35 Duties			
EO 004.01 Perform as a member of the ERT	Interpreting technical references	Date	Name and Sign
	Perform IAW HQ1 direction	Date	Name and Sign
	Clarify reports to HQ1	Date	Name and Sign
EO 004.02 Perform as the 35 Bulkhead Watch Keeper	Operating systems forward of 35 bulkhead	Date	Name and Sign
	Conducting rounds forward of 35 bulkhead	Date	Name and Sign
	Performing Emergency Operating Procedures (EOPs) as required	Date	Name and Sign

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Annex B - EQ2 W Eng Tech Submarine Fire Control

PO 005 Complete Final Board and Completion of Admin			
EO 006.01 Final Board	Section 1 Part A	Date	Name and Sign
	Section 1 Part B	Date	Name and Sign
	Section 2 Part A	Date	Name and Sign
	Section 2 Part B	Date	Name and Sign
	MMTT Completed	Date	Name and Sign
EO 006.02 Completion Cert	Completion Certificate Annex H Completed	Date	Name and Sign
EO 006.03	Complete Overall EQ Package in MMTT	Date	Name and Sign
EO 006.04 Message Released	Message Released Annex J Completed	Date	Name And Sign

Annex C – Weapons Engineering Technician Submarine SONAR

MARPAC RDIMS #1426099
Approved 02 Nov 2023

Annex C - EQ2 W Eng Tech Submarine SONAR

Introduction / Instructions

General Instructions

1. One copy of this EQ Annex will be provided to the member and one copy of Annex hereinafter referred to as the master copy, shall be made up for each individual beginning OJT and be retained by the Departmental Training Petty Officer.

2. A divisional interview shall be conducted to officially start the timeline and outline the process to the member. This is also an opportunity for the supervisor to agree to provide the member time to progress and complete training. The Unit / Member Agreement Form, at the end of these instructions, is to be signed during the divisional interview and kept with the Division Training PO. A copy is to be kept in the Annex as a reference to the member.

3. The tables of contents below include a list of equipment for you to gain experience on to make to prepared to become a Trade Journeyman. You will be required to identify the major components of the systems, how to operate the systems, perform Maintenance, and Perform a Mini Board on the system.

4. Each system below will have associated tables. On the right side of the tables are the date completed and Name and Signature blocks. The Name and Signature will be from the senior section technician or a W Eng Tech PO1. The PO's do not have to be completed in order and Name and Signatures can be granted through out the period. After the Name and Signatures are completed for each PO, a mini board is to be conducted and chaired by the CHOD or their designated PO1.

5. The development of the Member through the EQ process is critical to the creation of skilled and experienced technicians. To this end, the Name and Signature of the supervisors indicating completion of EOs and POs must reflect a high standard of accomplishment by the Member;

6. If deemed essential by the Sr Technician or Manager, the Member may have to perform more than the required number of times specified IOT satisfactorily complete the EO/PO at the high level of proficiency expected and get the Name and Signature.

Mini-Board Instructions

7. The board is made up of:

   a. CSED CHOD (scoring);

   b. SECTION HEAD (scoring);

   c. Subject Matter Expert (non-scoring member); and

8. Material and facilities requirement:

a. Pens, pencils and stationary;

b. Specified References;

c. Print out of marking guide (Annex G) for each member of the scoring board members; and

d. Suitable location.

9. Board Procedure:

a. Board members will meet with the member to ensure all supplies, references and equipment is available. They will explain the board procedures.

b. The member is then given 45 minutes before the Oral Board commencement to prepare the drawing as fitted in the member's ship.

c. The drawing must include:

i. Major Components;

ii. Major interconnections;

iii. Power Requirements; and

iv. Environmental requirements.

d. The member is complete the Oral Presentation in 25 Minutes by:

i. Describing the Purpose of system;

ii. Identify and explain major components;

iii. Identify and explain major signals flow; Identify and explain power requirements;

iv. Identify and explain environmental requirements;

v. Identify and talk about safety requirements; and

vi. Answer board questions.

10. Following the board, the member will be debriefed on their score, boardman-ship, strength, and areas for development.

<u>Final Board Instructions</u>

Annex C - EQ2 W Eng Tech Submarine SONAR

11. When the member has completed the associated Annex, the member can prepare for and then complete the Final Board. Details and instructions for the final board can be found at Annex F Final Board.

Administration Instructions

1. The time to complete and instruction for extensions are in the main instructions of the EQ.

12. Members are to be cautioned that failure to complete the EQ, including requests for Cease Training, may result in an Administrative Review IAW NPTORD 4500-29 Course Non-Concurrence, Deferrals and Removals.

13. As the member completes each EO and PO to the required standard, the Training PO shall record the progress of the member in MM, provide the Member with a copy and report the progress to the CHOD on a Monthly basis. The Progression Review Table is to be signed monthly by the Division Training PO and kept in their file with the Member Agreement Form.

14. It may not be possible to conduct some EOs/POs. If, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable, therefore:

c. The Departmental Training PO shall arrange an alternate unit/training establishment to conduct the EO/PO. Maintenance Procedural Trainer (MPT) may be utilized as a last resort if no other option is available within the timeframe of the EQ.

15. Name and Signatures may not be required under some PO/EO's. Some EO's will not apply, and N/A can be written into the Name and Signature block. For example, N/A can be written into the Confined Space Name and Signature block under Cranes.

Annex C - EQ2 W Eng Tech Submarine SONAR

Unit / Member Agreement Form

Name and Initial: _____

Rank: _____

SN: _____

Date Training Started: _____

Expected Date of Completion: _____

Divisional Interview Conducted by:

DivO: _____

Supervisor: _____

Note: THE EQ MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE EQ WILL BE RETURNED FOR RE-SUBMISSION.

Annex C - EQ2 W Eng Tech Submarine SONAR

PROGRESSION REVIEW

Month	EQ REVIEWED BY (RANK, POSITION, NAME)	SIGNATURE	DATE
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
First Extension (Unit CO)			
Second Extension (DNP 3)			

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Annex C - EQ2 W Eng Tech Submarine SONAR

Table of Contents

PO 001 Major SONAR Systems
EO 001.01 Bow Array
EO 001.01.01 Identify the Bow Array Major Components
EO 001.01.02 Operate the Bow Array
EO 001.01.03 Perform Maintenance on Bow Array
EO 001.02 Passive Ranging SONAR
EO 001.02.01 Identify the Passive Ranging Sonar Major Components
EO 001.02.02 Operate the Passive Ranging Sonar
EO 001.02.03 Perform Maintenance on Passive Ranging Sonar
EO 001.03 Flank Array
EO 001.03.01 Identify the Flank Array Major Components
EO 001.03.02 Operate the Flank Array
EO 001.03.03 Perform Maintenance on Flank Array
EO 01.04 Sail Array
EO 001.04.01 Identify the Sail Array Major Components
EO 001.04.02 Operate the Sail Array
EO 001.04.03 Perform Maintenance on Sail Array
EO 001.05 Active Intercept and Ranging System
EO 001.05.01 Identify the Active Intercept and Ranging System Major Components
EO 001.05.02 Operate the Active Intercept and Ranging System
EO 001.05.03 Perform Maintenance on Active Intercept and Ranging System
EO 001.06 Towed Array
EO 001.06.01 Identify the Towed Array Major Components
EO 001.06.02 Operate the Towed Array
EO 001.06.03 Perform Maintenance on Towed Array
EO 001.07 Own Ship Noise Monitoring System
EO 001.07.01 Identify the Own Ship Noise Monitoring System Major Components
EO 001.07.02 Operate the Own Ship Noise Monitoring System
EO 001.07.03 Perform Maintenance on Own Ship Noise Monitoring System
EO 001.08 Perform Mini Board on Major SONAR Systems

PO 002 Minor SONAR Systems
EO 002.01 Echo Sounder
EO 002.01.01 Identify the Echo Sounder Major Components
EO 002.01.02 Operate the Echo Sounder
EO 002.01.03 Perform Maintenance on Echo Sounder
EO 002.02 Sonar 2004 Bathy
EO 002.02.01 Identify the -Sonar 2004 Bathy Major Components
EO 002.02.02 Operate the -Sonar 2004 Bathy
EO 002.02.03 Perform Maintenance on -Sonar 2004 Bathy
EO 002.03 Mk 48 Bathythermograph
EO 002.03.01 Identify the Mk 8F Bathythermograph Major Components
EO 002.03.02 Operate the Mk 8F Bathythermograph
EO 002.03.03 Perform Maintenance on Mk 8F Bathythermograph

Annex C - EQ2 W Eng Tech Submarine SONAR

EO 002.04 SONAR 2008 Underwater Telephone
EO 002.04.01 Identify the Sonar 2008 Underwater Telephone Major Components
EO 002.04.02 Operate the Sonar 2008 Underwater Telephone
EO 002.04.03 Perform Maintenance on Sonar 2008 Underwater Telephone
EO 002.05 SONAR 2073 Emergency Underwater Telephone
EO 002.05.01 Identify the Sonar 2073 Emergency Underwater Telephone Major Components
EO 002.05.02 Operate the Sonar 2073 Emergency Underwater Telephone
EO 002.05.03 Perform Maintenance on Sonar 2073 Emergency Underwater Telephone
EO 003.06 Sonar 2071 Noise Augmentation System
EO 002.06.01 Identify the Sonar 2071 Noise Augmentation System Major Components
EO 002.06.02 Operate the Sonar 2071 Noise Augmentation System
EO 002.06.03 Perform Maintenance on Sonar 2071 Noise Augmentation System
EO 002.07 Perform Mini Board on Minor SONAR Systems

PO 003 Navigation Aids
EO 003.01 SINS Ring Laser Gyro Navigator
EO 003.01.01 Identify the SINS Ring Laser Gyro Navigator Major Components
EO 003.01.02 Operate the SINS Ring Laser Gyro Navigator
EO 003.01.03 Perform Maintenance on SINS Ring Laser Gyro
EO 003.02 Navigator Navigation Data Distribution System (NDDS)
EO 003.02.01 Identify the NDDS Major Components
EO 003.02.02 Operate the NDDS
EO 003.02.03 Perform Maintenance on NDDS
EO 003.03 Automatic Identification System (AIS)
EO 003.03.01 Identify the AIS Major Components
EO 003.03.02 Operate the AIS
EO 003.03.03 Perform Maintenance on AIS
EO 003.04 Shinnads Dual Monitor (SDM)
EO 003.04.01 Identify the SDM Major Components
EO 003.04.02 Operate the SDM
EO 003.04.03 Perform Maintenance on SDM
EO 003.05 AGILOG, Ship Speed and Distance System
EO 003.05.01 Identify the AGILOG Major Components
EO 003.05.02 Operate the AGILOG
EO 003.05.03 Perform Maintenance on AGILOG
EO 003.06 Log RTU System
EO 003.06.01 Identify the Log RTU System Major Components
EO 003.06.02 Operate the Log RTU
EO 003.06.03 Perform Maintenance on Log RTU
EO 003.07 Depth Sounder
EO 003.07.01 Identify the Depth Sounder Major Components
EO 003.07.02 Operate the Depth Sounder
EO 003.07.03 Perform Maintenance on Depth Sounder
EO 003.08 Navigation Information Display System (NIDS)
EO 003.08.01 Identify the NIDS and Ship’s Attitude Panels Major Components

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EO 003.08.02 Operate the NIDS and Ship's Attitude Panels
EO 003.08.03 Perform Maintenance on NIDS and Ship's Attitude Panels
EO 003.09 Perform Mini Board on Navigation Aids

PO 004 Periscopes Systems
EO 004.01 Attack Periscope
EO 004.01.01 Identify the Attack Periscope Major Components
EO 004.01.02 Operate the Attack Periscope
EO 004.01.03 Perform Maintenance on Attack Periscope
EO 004.02 Search Periscope
EO 004.02.01 Identify the Search Periscope Major Components
EO 004.02.02 Operate the Search Periscope
EO 004.02.03 Perform Maintenance on Search Periscope
EO 004.03 Tactical Television Console
EO 004.03.01 Identify the Tactical Television Console, TTVC or Periscope Control System, PCS Major Components
EO 004.03.02 Operate the Tactical Television Console, TTVC or Periscope Control System, PCS
EO 004.03.03 Perform Maintenance on Tactical Television Console, TTVC or Periscope Control System, PCS
EO 004.04 TMCU
EO 004.04.01 Identify the TMCU Major Components
EO 004.04.02 Operate the TMCU
EO 004.04.03 Perform Maintenance TMCU
EO 004.05 Perform Mini Board on Periscopes Systems

PO 005 Submarine Fire Control System
EO 005.01 CCS876
EO 005.01.01 Identify the CCS876 Major Components
EO 005.01.02 Operate the CCS876
EO 005.01.03 Perform Maintenance on CCS876
EO 005.02 Multi Tube Weapon Simulator (MTWS)
EO 005.02.01 Identify the MTWS Major Components
EO 005.02.02 Operate the MTWS
EO 005.02.03 Perform Maintenance on MTWS
EO 005.03 Weapon System Data Bus
EO 005.03.01 Identify the Weapon System Data Bus Major Components
EO 005.03.02 Operate the Weapon System Data Bus
EO 005.03.03 Perform Maintenance on Weapon System Data Bus
EO 005.04 Weapons Control Module (WCM)
EO 005.04.01 Identify the WCM Major Components
EO 005.04.02 Operate the WCM
EO 005.04.03 Perform Maintenance on WCM
EO.005.05 Perform Mini Board on Submarine Fire Control System

PO 006 ERT and Bulkhead 35 Duties

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EO 006.01 Perform as a member of the ERT
EO 006.02 Perform as the 35 Bulkhead Watch Keeper

PO 007 Final Board

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 001 Major SONAR Systems			
EO 001.01.01 Identify the Bow Array Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.01.02 Operate the Bow Array		Date	Signature
Operate the Bow Array	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		

EO 001.01.03 Perform Maintenance on Bow Array		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 001 Major SONAR Systems			
EO 001.02.01 Identify the Passive Ranging Sonar Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.02.02 Operate the Passive Ranging Sonar		Date	Signature
Operate the Passive Ranging SONAR	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		

EO 001.02.03 Perform Maintenance on Passive Ranging Sonar		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 001 Major SONAR Systems			
EO 001.03.01 Identify the Flank Array Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.03.02 Operate the Flank Array		Date	Signature
Operate the Flank Array	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		

Annex C - EQ2 W Eng Tech Submarine SONAR

EO 001.03.03 Perform Maintenance on Flank Array		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 001 Major SONAR Systems			
EO 001.04.01 Identify the Sail Array Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.04.02 Operate the Sail Array		Date	Signature
Operate the Sail Array	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		

Annex C - EQ2 W Eng Tech Submarine SONAR

EO 001.04.03 Perform Maintenance on Sail Array		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 001 Major SONAR Systems			
EO 001.05.01 Identify the Active Intercept and Ranging System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.05.02 Operate the Active Intercept and Ranging System		Date	Signature
	Perform 48 hour pre-sails		

Operate the Active Intercept and Ranging System	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 001.05.03 Perform Maintenance on Active Intercept and Ranging System		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 001 Major SONAR Systems			
EO 001.06.01 Identify the Towed Array Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.06.02 Operate the Towed Array		Date	Signature
	Perform 48 hour pre-sails		

Operate the Towed Array	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 001.06.03 Perform Maintenance on Towed Array		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 001 Major SONAR Systems			
EO 001.07.01 Identify the Own Ship Noise Monitoring System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 001.07.02 Operate the Own Ship Noise Monitoring System		Date	Signature
	Perform 48 hour pre-sails		

Operate the Own ship Noise Mont System	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 001.07.03 Perform Maintenance on Own Ship Noise Monitoring System		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

<table>
<tr><th colspan="4">EO 001.08 Perform Mini Board on Major SONAR Systems</th></tr>
<tr><td rowspan="6">Draw system block diagram to include</td><td>Major Components</td><td>Date Completed</td><td>Signature</td></tr>
<tr><td>Major Signals Flow</td><td></td><td></td></tr>
<tr><td>Power requirements</td><td colspan="2">Board Member Chair name</td></tr>
<tr><td>Environmental requirements</td><td colspan="2" rowspan="2"></td></tr>
<tr><td>Environmental requirements</td></tr>
<tr><td rowspan="7">Explain / talk about</td><td>Purpose of system</td><td colspan="2">Overall Board Comments</td></tr>
<tr><td>Identify and explain major components</td><td colspan="2" rowspan="6"></td></tr>
<tr><td>Identify and explain major signals flow</td></tr>
<tr><td>Identify and explain power requirements</td></tr>
<tr><td>Identify and explain environmental requirements</td></tr>
<tr><td>Identify and talk about safety requirements</td></tr>
</table>

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PO 002 Minor SONAR Systems			
EO 002.01.01 Identify the Echo Sounder Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.01.02 Operate the Echo Sounder		Date	Signature
Operate Equipment	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		

EO 002.01.03 Perform Maintenance on Echo Sounder		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 002 Minor SONAR Systems			
EO 002.02.01 Identify the -Sonar 2004 Bathy Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.02.02 Operate the -Sonar 2004 Bathy		Date	Signature
Operate the Equipment	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		

EO 002.02.03 Perform Maintenance on -Sonar 2004 Bathy		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 002 Minor SONAR Systems			
EO 002.03.01 Identify the Mk 8F Bathythermograph Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.03.02 Operate the Mk 8F Bathythermograph		Date	Signature
	Perform 48 hour pre-sails		

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Operate the Equipment	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 002.03.03 Perform Maintenance on Mk 8F Bathythermograph		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 002 Minor SONAR Systems			
EO 002.04.01 Identify the Sonar 2008 Underwater Telephone Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.04.02 Operate the Sonar 2008 Underwater Telephone		Date	Signature
	Perform 48 hour pre-sails		

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Operate the Equipment	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 002.04.03 Perform Maintenance on Sonar 2008 Underwater Telephone		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 002 Minor SONAR Systems			
EO 002.05.01 Identify the Sonar 2073 Emergency Underwater Telephone Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.05.02 Operate the Sonar 2073 Emergency Underwater Telephone		Date	Signature
	Perform 48 hour pre-sails		

Operate the Equipment	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 002.05.03 Perform Maintenance on Sonar 2073 Emergency Underwater Telephone		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 002 Minor SONAR Systems			
EO 002.06.01 Identify the Sonar 2071 Noise Augmentation System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 002.06.02 Operate the Sonar 2071 Noise Augmentation System		Date	Signature
	Perform 48 hour pre-sails		

Operate the Equipment	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 002.06.03 Perform Maintenance on Sonar 2071 Noise Augmentation System		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

EO 002.07 Perform Mini Board on Minor SONAR Systems			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

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PO 003 Navigation Aids			
EO 003.01.01 Identify the SINS Ring Laser Gyro Navigator Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.01.02 Operate the SINS Ring Laser Gyro Navigator		Date	Signature

Operate the SINS	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.01.03 Perform Maintenance on SINS Ring Laser Gyro		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 003 Navigation Aids			
EO 003.02.01 Identify the Navigator Navigation Data Distribution System (NDDS) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.02.02 Operate the NDDS		Date	Signature
	Perform 48 hour pre-sails		

Operate the NDDS	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.02.03 Perform Maintenance on NDDS		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 003 Navigation Aids			
EO 003.03.01 Identify the Automatic Identification System (AIS) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.03.02 Operate the AIS		Date	Signature
	Perform 48 hour pre-sails		

Operate the AIS	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.03.03 Perform Maintenance on AIS		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 003 Navigation Aids			
EO 003.04.01 Identify the Shinnads Dual Monitor (SDM) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.04.02 Operate the SDM		Date	Signature
	Perform 48 hour pre-sails		

Operate the SDM	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.04.03 Perform Maintenance on SDM		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 003 Navigation Aids			
EO 003.05.01 Identify the AGILOG, Ship Speed and Distance System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.05.02 Operate the AGILOG		Date	Signature
	Perform 48 hour pre-sails		

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Operate the AGILOG	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.05.03 Perform Maintenance on AGILOG		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 003 Navigation Aids			
EO 003.06.01 Identify the Log RTU System Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.06.02 Operate the Log RTU		Date	Signature

Operate the RTU	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.06.03 Perform Maintenance on Log RTU		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 003 Navigation Aids			
EO 003.07.01 Identify the Depth Depth Sounder Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.07.02 Operate the Depth Sounder		Date	Signature
	Perform 48 hour pre-sails		

Operate the Depth Sounder	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.07.03 Perform Maintenance on Depth Sounder		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 003 Navigation Aids			
EO 003.08.01 Identify the Navigation Information Display System, NIDS and Ship's Attitude Panels Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 003.08.02 Operate the Navigation Information Display System, NIDS and Ship's Attitude Panels		Date	Signature
	Perform 48 hour pre-sails		

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Operate the Equipment	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 003.08.03 Perform Maintenance on Navigation Information Display System, NIDS and Ship's Attitude Panels		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

<table>
<tr><th colspan="4">EO 003.09 Perform Mini Board on Navigation Aids</th></tr>
<tr><td rowspan="6">Draw system block diagram to include</td><td>Major Components</td><td>Date Completed</td><td>Signature</td></tr>
<tr><td>Major Signals Flow</td><td></td><td></td></tr>
<tr><td>Power requirements</td><td colspan="2">Board Member Chair name</td></tr>
<tr><td>Environmental requirements</td><td colspan="2" rowspan="2"></td></tr>
<tr><td>Environmental requirements</td></tr>
<tr><td rowspan="7">Explain / talk about</td><td>Purpose of system</td><td colspan="2">Overall Board Comments</td></tr>
<tr><td>Identify and explain major components</td><td colspan="2" rowspan="6"></td></tr>
<tr><td>Identify and explain major signals flow</td></tr>
<tr><td>Identify and explain power requirements</td></tr>
<tr><td>Identify and explain environmental requirements</td></tr>
<tr><td>Identify and talk about safety requirements</td></tr>
</table>

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PO 004 Periscopes Systems			
EO 004.01.01 Identify the Attack Periscope Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 004.01.02 Operate the Attack Periscope		Date	Signature
	Perform 48 hour pre-sails		

Operate the Attack Periscope	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 004.01.03 Perform Maintenance on Attack Periscope		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 004 Periscopes Systems			
EO 004.02.01 Identify the Search Periscope Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 004.02.02 Operate the Search Periscope		Date	Signature
	Perform 48 hour pre-sails		

Operate the Search Periscope	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 004.02.03 Perform Maintenance on Search Periscope		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

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PO 004 Periscopes Systems			
EO 004.03.01 Identify the Tactical Television Console, TTVC or Periscope Control System, PCS Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 004.03.02 Operate the Tactical Television Console, TTVC or Periscope Control System, PCS		Date	Signature

Operate the Equipment	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 004.03.03 Perform Maintenance on Tactical Television Console, TTVC or Periscope Control System, PCS		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 004 Periscopes Systems			
EO 004.04.01 Identify the TMCU Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 004.04.02 Operate the TMCU		Date	Signature
	Perform 48 hour pre-sails		

Operate the TMCU	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 004.04.03 Perform Maintenance TMCU		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

EO 004.05 Perform Mini Board on Periscopes Systems			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

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Annex C - EQ2 W Eng Tech Submarine SONAR

PO 005			
EO 005.01.01 Identify the CCS876 Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 005.01.02 Operate the CCS876		Date	Signature
	Perform 48 hour pre-sails		

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Operate the CCS876	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 005.01.03 Perform Maintenance on CCS876		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 005			
EO 005.02.01 Identify the Multi Tube Weapon Simulator (MTWS) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 005.02.02 Operate the MTWS		Date	Signature
	Perform 48 hour pre-sails		

Operate the MTWS	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 005.02.03 Perform Maintenance on MTWS		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 005			
EO 005.03.01 Identify the Weapon System Data Bus Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 005.03.02 Operate the Weapon System Data Bus		Date	Signature

Operate the Weapon System Data Bus	Perform 48 hour pre-sails		
	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 005.03.03 Perform Maintenance on Weapon System Data Bus		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 005			
EO 005.04.01 Identify the Weapons Control Module (WCM) Major Components		Date	Signature
Identify Major Components	Name of component		
	Purpose of component		
Identify Environmental requirements	Identify required environmental for normal operation		
	Identify Emergency Environmental		
Identify Power requirements	Identify Safety precautions for power		
	Identify 120V		
	Identify 240V		
	Identify 400 HZ		
	24V Grade 1		
	24V Grade 2		
Identify Key requirements	Transmit Keys		
	Firing Keys		
	Enabling Keys		
Identify safety switches	Man-a-loft		
	Maintenance		
	Interlocks		
Identify miscellaneous hazards	Hydraulics		
	Pneumatics		
	RADHAZ		
	HAZMAT		
	Explosives		
	Confined space		
	Watertight Integrity		
EO 005.04.02 Operate the WCM		Date	Signature
	Perform 48 hour pre-sails		

Operate the WCM	Perform Flash up		
	Verify operational		
	Brief SECTION HEAD		
EO 005.04.03 Perform Maintenance on WCM		Date	Signature
Assist with / Perform Minor PM	Identify and explain CFTO (NY's)		
	Asist with Daily PM Routine		
	Asist with Weekly PM Routine		
	Asist with Monthly PM's Routine		
Assist with Major PM	Asist with 6 Monthly PM Routine		
	Asist with 12 Monthly PM Routine		
	Assist with 24 Monthly PM Routine *See Note 1		
	Assist with 48 Monthly PM Routine *See Note 1		
Miscellaneous Maintenance	Assist with Performance Tests *See Note 1		
	Assist with Trials *See Note 1		
	Assist with TB's *See Note 1		
Assist with CM	Assist with identifying faulty LRU		
	Assist with Replacing Faulty LRU		
	Assist with verifying system operational		

Note 1: If unable to complete the Routine, the member is to, with a trained technician, read over the routine and conduct a walk through to the satisfaction of the Senior Hand.

Annex C - EQ2 W Eng Tech Submarine SONAR

EO 005.05 Perform Mini Board on Submarine Fire Control System			
Draw system block diagram to include	Major Components	Date Completed	Signature
	Major Signals Flow		
	Power requirements	Board Member Chair name	
	Environmental requirements		
	Environmental requirements		
Explain / talk about	Purpose of system	Overall Board Comments	
	Identify and explain major components		
	Identify and explain major signals flow		
	Identify and explain power requirements		
	Identify and explain environmental requirements		
	Identify and talk about safety requirements		

Note 1: If unable to complete because Routine did not come up during training period, the member is to, with a trained technician, read over the routine and conduct a walk through.

Annex C - EQ2 W Eng Tech Submarine SONAR

PO 006 ERT and Bulkhead 35 Duties			
EO 006.01 Perform as a member of the ERT	Interpreting technical references	Date	Name and Sign
	Perform IAW HQ1 direction	Date	Name and Sign
	Clarify reports to HQ1	Date	Name and Sign
Perform as the 35 Bulkhead Watch Keeper	Operating systems forward of 35 bulkhead	Date	Name and Sign
	Conducting rounds forward of 35 bulkhead	Date	Name and Sign
	Performing Emergency Operating Procedures (EOPs) as required	Date	Name and Sign

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Annex C - EQ2 W Eng Tech Submarine SONAR

PO 007 Final Board			
EO 006.01 Final Board	Section 1 Part A	Date	Name and Sign
	Section 1 Part B	Date	Name and Sign
	Section 2 Part A	Date	Name and Sign
	Section 2 Part B	Date	Name and Sign
	MMTT Completed	Date	Name and Sign
EO 006.02 Completion Cert	Completion Certificate Annex H Completed	Date	Name and Sign
EO 006.03	Complete Overall EQ Package in MMTT	Date	Name And Sign
EO 006.04 Message Released	Message Released Annex J Completed	Date	Name And Sign

Annex F – Final Board

Annex F – Final Board

1. The final board is broken into 2 sections with 2 parts in each section:

    a. Section 1:

        i. Part A: Suite Board; and

        ii. Part B: Sub-Suite Board.

    b. Section 2:

        i. Part A: Sub-Suite Flash-up; and

        ii. Part B: Suite Walk through.

<u>Section 1 Part A / Part B Board presentations</u>

1. Section 1 is broken into 2 boards to be completed one after the other. First, for the overall Suite that the member has been assigned as a sub occupation of the W Eng Tech trade, and then the board of a Sub-Suite.

2. For both parts, the board is made up of:

    a. CSED HOD (scoring) (Chairperson);

    b. CSED CHOD (scoring);

    c. STOW (scoring);

    d. Subject Matter Expert (non-scoring member); and

    e. Honest Broker (non-scoring member). The Honest Broker shall monitor the board for consistence, conduct and impartiality of the board members. The FCSEC should be to be the honest broker if available. If they are unavailable, the name, rank, and trade of suggested honest broker should be approved by the FCSEC prior to the board.

3. Material and facilities requirement:

    a. Pens, pencils, dry erase markers, ruler, whiteboard and stationary;

    b. Print out of marking guide (Annex G) for each member of the scoring board members (2 copies for each member); and

c. Suitable location.

Board Procedure:

1. Board members will meet with the member to ensure all supplies are available. They will then explain the board procedures and assign which Sub-Suite is to be presented. The Sub-Suite will be selected IAW Table 1 Sub-Suite Selections.

2. The member is then given 60 minutes before the Oral Boards commencement to prepare the drawings (both the Suite and Sub-Suite drawings) as fitted in the member's ship. An example of the Board layout is located at Table 2 Board Layout example.

3. On completion of the drawing time, the board members are to assemble at the location and the Chairperson (HOD) is to indicate when Part A of the board is to commence.

4. The member is to complete the Oral Presentation in 25 Minutes IAW the guidelines listed below.
5. On completion of Part A, the member is to answer questions from the board.

6. The Second Oral Presentation should not commence until:

   a. The Chairperson (HOD) indicates the first board is complete; and

   b. The member has been deemed Satisfactory IAW the Marking Guide by all the scoring members. If the first presentation was deemed not satisfactory, the Finial Board is not to proceed any further at this time. Unsuccessful board guidelines are found further in Annex A.

Part A Guidelines:

1. The drawing must include:

   a. The name of the 4 PO's (exclude the PO for Part B);

   b. The Sub-systems of each PO;

   c. List of subjects to be covered (member may add more):

      i. Name of PO;
      ii. Name of sub-system;
      iii. Purpose of sub-system;
      iv. Safety considerations; and
      v. Location of equipment.

Annex F – Final Board

Part B Guidelines:

2. The drawing must include:

   a. The name of the PO (assigned at initial meeting);

   b. The Sub-systems of the PO;

   c. List of subjects to be covered (member may add more):

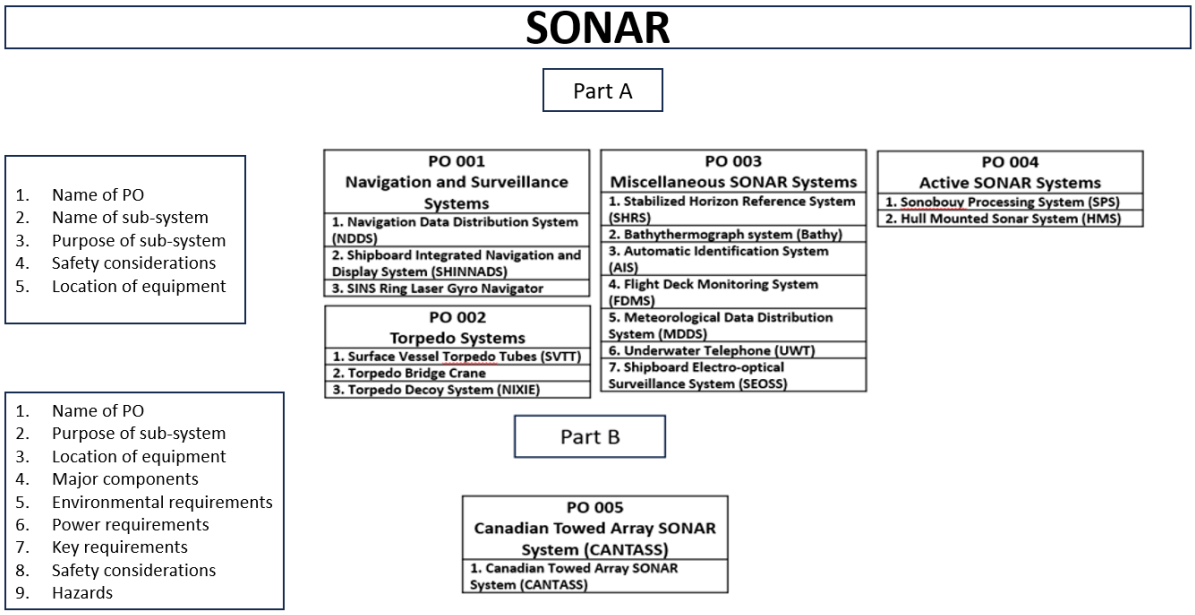
      i. Name of PO;
      ii. Purpose of sub-system;
      iii. Location of equipment;
      iv. Major components;
      v. Environmental requirements;
      vi. Power requirements;
      vii. Key requirements;
      viii. Safety considerations; and
      ix. Hazards.

Table 1- Sub-Suite Selections

W Eng Tech Communications	PO 003 HF, VHF, and UHF Communications
	PO 004 Protected MILSATCOM Systems
	PO 005 RADAR Suite
W Eng Tech Fire Control	PO 001 Weapons Handling System
	PO 002 Weapons Discharge System
	PO 003 Submerged Signal Ejector
W Eng Tech SONAR	PO 001 Major SONAR Systems
	PO 002 Minor SONAR Systems
	PO 005 Submarine Fire Control System

Annex F – Final Board

Table 2 - Board Layout example



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Annex F – Final Board

Section 2

1. Section 2 Boards do not have to completed in any order. For Example, Part A Flash-up can be conducted at a time when the ship is conducting a Flash-up to sail. Part B can be conduct prior to Part A if decided. Both Boards should be completed as close to each other as practical, but to be determined by the unit.

Section 2 Part A: Flash-up, initialize, verify it is operational or report if faulted.

3. For both parts, the board is made up of:

    f. CSED HOD (scoring) (Chairperson);

    g. CSED CHOD (scoring);

    h. SECTION HEAD (scoring); and

4. Material and facilities requirement:

    a. Pens, pencils and stationary;

    b. Specified References; and

    c. Print out of marking guide (Annex G) for each member of the scoring board (1 copy for each Part).

4. The Sub-Suite to be flashed-up is to be chosen by chair and identified to the Board members ahead of time. The sub-suite to be chosen is to be selected from Table 1 Sub-Suite Selections.

5. The Board members are to ensure the following is completed prior to the commencement of the flash-up:

    a. Minor faults (minor faults are items like switches in wrong position or a Circuit braker turned off. No faults are to be placed on the equipment) are in place and noted by the Board;

    b. Required Spaces are unlocked ahead of time;

    c. Units Flash-up checklist to be used by member is to be printed; and

    d. Keys are drawn by Board members ahead of time. (The Member is not to be tested on procedures to draw Keys on this flash-up. Flash-up keys are normally drawn by the Flash-up I/C and handed out).

6. The board and the member are to meet in the CSE Office prior to the flash-up commencing and the following is to take place:

   a. The Sub-Suite to be flashed-up is to be made know to the member;

   b. Keys are to be identified to the member but held by the Board members until asked for them by the member;

   c. The Flash-up check list, pen/pencil and clip board are to be given to the member;

   d. The opportunity for the member to ask questions is to be given; and

   e. The time limit of 1 hour is to be marked at the start of the flash-up.

7. To be successful the member is to:

   a. Complete the flash-up in 1 hour;

   b. Locate the spaces and components to be flashed-up;

   c. Turn-on the system IAW the manuals. The SME is to monitor the member to ensure the flash-up is done correctly. The member is to be stopped immediately if procedures are not being correctly followed or if there is a safety concern;

   d. The system is to be considered flashed-up when the equipment is at the state considered to be operation IAW the units pre-sail flash-up check list. IE: if flashing the Sub-Suite RADAR – Air Search RADAR, there will be no requirement to radiate the system out the antenna;

   e. Report status of system to board members; and

   f. If the flash-up was deemed not successful, the Finial Board is not to proceed any further at this time. Unsuccessful board guidelines are found further in Annex F.

Section 2 Part B: Is the Suite Walk Through.

1. The tour is to be led by the member. Upon entering a space, the member is to:

   a. Identify the major components;

   b. Identify any Man-aloft key locations;

   c. Identify any Hazards;

d. Identify any Safety switches; and

e. Identify any Environmental concerns.

Board Administration

Unsuccessful board guidelines

If unsuccessful:

1. Conduct 2nd attempt for the unsatisfactory part. It will not be necessary to restart the entire section.

2. If 2nd attempt is unsuccessful, then:

    a. The Occupation Manager is to be notified;

    b. A meeting is to be conducted to include (as many as possible. At a minimum, the OM, TM, and divisional representation are required).

        i. Occupation Manager;

        ii. Training Manager;

        iii. Career Manager;

        iv. Coastal Advisor;

        v. Coastal Trade manager; and

        vi. The member's HOD, CHOD and DivO.

    c. The members of the meeting will decide:

        i. For the member to attempt the Final Board again;

        ii. Recommend that an A/R be conducted by the CM; or

        iii. Recommend a new Sub-trade of the W Eng Tech be assigned and a new EQ package is to be assigned.

Successful board guidelines

3. Undo completion of Section 1 and Section 2, the **PO Complete Final Board and Completion of Admin** is to be completed.

Annex G – Marking Guide

Annex G – Marking Guide

W Eng Tech Experience Qualification Final Board MOSID RANK QUALIFICATION BOARD (AKNV) MARKING GUIDE	
Name of Member:	Unit:
SN	Date:
CSEO (Chairperson):	CSEC:
SECTION HEAD:	Honest Broker:
SME:	
Identify which part of the final board: (Circle one). A new Marking Guide to be used for each task.	1. Section 1 Part A: Suit Board; 2. Section 1 Part B: Sub-Suite Board; 3. Section 2 Part A: Sub-Suite Flash-up; or 4. Section 2 Part B: Suite Walk through.
The second sheet of the marking guide is a remarks/comments/notes area to be used by the board members while the member is conducting the task. All marking guides from all board members are to be kept until all sections of the final board is complete.	
On completion of the board, circle the number on the left IAW the description on the right. 5 is the minimum stand required to pass.	
0	Cannot even begin to complete the task.
1	Never studied but knows from who/where more information can be obtained to complete the task
2	Very low knowledge of topic–only able to complete part of the task.
3	Low knowledge: needs continuous prodding and assistance to complete the task.
4	Minor gaps in knowledge. Unable to discuss points expected. Does not prove competent to complete task on own.
5 Min standard to pass	Satisfactory but needs minor prodding or assistance in answering the main questions and to complete the task.
6	Satisfactory: Answers most questions in acceptable detail; completes task.
7	Answers questions comprehensively. Follow-up questions are answered in detail. Overall demonstrates a strong understanding of the task and completes task with no problems.
8	Answers questions completely and elaborates, in detail on specific aspects such that only a few follow-up questions are necessary; Task completed with confidence.

Annex G – Marking Guide

W Eng Tech Experience Qualification Final Board MOSID RANK QUALIFICATION BOARD (AKNV) MARKING GUIDE	
Name of Member:	Unit:
SN	Date:
CSEO (Chairperson):	CSEC:
SECTION HEAD:	Honest Broker:
SME:	
Identify which part of the final board: (Circle one). A new Marking Guide to be used for each task.	1. Section 1 Part A: Suit Board; 2. Section 1 Part B: Sub-Suite Board; 3. Section 2 Part A : Sub-Suite Flash-up; or 4. Section 2 Part B: Suite Walk through.
remarks/comments/notes Area. Please write your comments and notes on the final board section here. Any question should be noted. Reason for the member not being successful shall also be noted here.	
Remarks/Comments/Notes	

Annex H– Completion Certificate

Annex H – Completion Certificate

CERTIFICATION OF EXPERIENCE QUALIFICATION

I certify that _____ has completed the requirements to the standards stated in the Weapons Engineering Technician EQ2 Annex:_____- MITE Code _____.

Member is to be granted the above qualification/Mite Code and the National Qualification AKYJ – MITE Code 123768.

HRMS update action be taken granting the AKYJ National Qualification and change member's MOSID to 00366-01 and apply specialist grade 1 pay effective the training completion date. This Annex shall be retained on the member's PERs file.

Signed IAW Naval Personnel Instructions (NPI)#13 Sep 2019

CHOD_____ DATE _____

(Print and sign name)

Annex J – Message

Annex J – Messages

1. Upon successful completion of the Final Board the convening authority will draft a message:

R DATE, TIME, GROUP

FM (HOME UNIT)
TO NDHQ DGMC OTTAWA// D MIL C 2-9-2//
INFO RCCPUVA/ NDHQ C NAVY DNP OTTAWA//DNP 3-2-3
NPTG HQ ESQUIMALT//
CANFLTPAC HQ ESQUIMALT// F4 ENG//
CANFLTLANT HQ HALIFAX// F4 ENG//

PROTECTED A
SIC WAB

SUBJ: WEAPONS ENGINEERING TECHNICIAN AKYJ EQ 2 COMPLETION

REF: AKYJ EQ JR W ENG TECH DATED 02NOV2023

4. FOL MBR HAS COMPLETED ALL REQ IAW REF AND IS GRANTED W ENG TECH JR AKYJ EQ NATIONAL QUALIFICATION READ IN 7 COLUMNS:

(SN/RANK/NAME/TRADE/MITE CODE/START DATE/COMPLETION DATE/REMARKS)

5. REQ UNIT HRMS UPDATE ACTION BE TAKEN TO GRANT AKYJ EFFECTIVE DDMMYYYY

6. POC DEPARTMENTAL TRAINING PETTY OFFICER, RANK, NAME, INITIALS, NAME (AT)FORCES.GC.CA PHONE XXX-XXX-XXXX

Annex J – Message

2. If the member does not complete the package for reason of Deferral /Non-Concurrence /Removal the follow must occur:

   a. Members are to be cautioned that failure to complete the EQ, including requests for Cease Training, may result in an Administrative Review IAW NPTORD 4500-29 Course Non-Concurrence, Deferrals and Removals.

   b. If a Deferral /Non-Concurrence /Removal is supported by the members CoC, the Deferral /Non-Concurrence /Removal Message (NPTORD 4500-29 Course Non-Concurrence, Deferrals and Removals - Annex B) is to be released to the following addresses:

      TO NDHQ DGMC OTTAWA// D MIL C 2-9-2//
      INFO RCCPUVA/ NDHQ C NAVY DNP OTTAWA//DNP 3-2-3
      NPTG HQ ESQUIMALT//W ENG TECH TM//
      CANFLTPAC HQ ESQUIMALT// F4 ENG//
      CANFLTLANT HQ HALIFAX// F4 ENG//